

Bill of Materials (BOM)

Industrial Energy Management System

Hardware Components

Sl. No.	Component	Specification	Quantity	Purpose
1	RuggedBoard A5D2x	ARM Cortex-A5 Embedded Development Board	1	Main controller for data acquisition, processing, MQTT communication, and relay control
2	ACS712 Current Sensor Module	20 A Analog Current Sensor	1	Measures load current
3	RS485 Modbus Relay Module	2-Channel, 5 V Relay Module	1	Controls the AC load remotely
4	AC Bulb	230 V AC	1	Demonstration load
5	Bulb Holder	230 V AC	1	Holds the demonstration bulb
6	Ethernet Cable	CAT5e/CAT6	1	Provides network connectivity between the RuggedBoard and the host system
7	Micro-USB Cable	USB to Micro-USB	1	Powers and programs the RuggedBoard
8	Hook-up Wires	Male-to-Male / Screw Terminal Wires	As required	Electrical connections between components
9	AC Plug with Power Cord	230 V AC	1	Supplies mains power to the load

Software Requirements

Software	Purpose
Linux	Operating system running on the RuggedBoard
C Programming Language	Embedded firmware development
GCC Toolchain	Compiling the firmware
MQTT (Mosquitto)	Telemetry communication
ThingsBoard	IoT dashboard and visualization
Git	Version control

Software	Purpose
GitHub	Project hosting and documentation

Communication Interfaces

Interface	Purpose
Ethernet	Network communication
MQTT	Telemetry transmission
RS485 (Modbus RTU)	Relay communication
ADC	Reading the analog output of the ACS712 sensor

Measured Parameters

Parameter	Source
Current (A)	ACS712 Current Sensor
Voltage (V)	Assumed constant at 230 V AC
Power (W)	Calculated from voltage and current
Energy (kWh)	Calculated by integrating power over time